1. Item 4 (Most Relevant)

Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

📌 **Why is this the most important?**
This problem primarily assesses whether students can:

- Correctly interpret "only open from 1 July to 27 August"
- Understand the meaning of "on average about how many people each day"

📌 **How it is reflected in the problem:**

- Students must translate **verbal information** into a mathematical situation
- They must recognise that "on average" implies **total number ÷ total number of days**
- Although the context is simple, accurate interpretation of the time span is essential

➡️ **Without correct reading and interpretation, mathematical modelling is impossible**

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2. Item 6

Identify and describe the mathematical aspects of a real-world problem situation, including identifying the significant variables

📌 **How this appears in the problem:**

- The problem is set in a real-world context (people climbing Mount Fuji)
- The underlying mathematical structure involves:
  - Number of people
  - Length of time
  - Average value

📌 **What students need to do:**

- Extract the relevant mathematical quantities
- Ignore non-essential contextual details (e.g. information about Mount Fuji itself)

3. Item 2

Identify the key variables in a model

📌 Key variables are clearly defined:

- Total number of climbers: $N = 200,000$
- Number of days: $T =$ days from 1 July to 27 August
- Average per day: $\frac{N}{T}$

📌 This step forms the bridge between context and calculation

4. Item 7

Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

📌 **Simplifications involved:**

- Assuming the number of climbers is **approximately evenly distributed** across days
- Ignoring variations due to weather, weekends, or holidays

📌 **Students must focus on the core structure:**

- Retain only the relationship **total ÷ time**

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5. Item 12

Identify the constraints, assumptions, and simplifications in a mathematical model

📌 **Implicit assumptions include:**

- Each day has roughly the same number of climbers
- The mountain is open every day in the given period
- "About 200,000" indicates an approximate total

📌 **This supports conceptual understanding of "average"**

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6. Item 9

Translate a problem into a standard mathematical representation or algorithm

📌 **The translation is straightforward:**

$$\text{Average per day} = \frac{200,000}{\text{number of days}}$$

📌 **This represents the shift from language to formula**

7. Item 8

Recognise aspects of a problem that correspond with known mathematical concepts, facts, or procedures

📌 **Relevant mathematical concepts:**

- Mean (average)
- Rate
- Quantity per unit time

📌 **This is a standard "average rate" problem**

8. Item 5

Recognise mathematical structure (including regularities, relationships, and patterns)

📌 **Recognised structure:**

- Linear relationship between total quantity, time, and unit rate
- Inverse relationship between average per day and number of days

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9. Item 1

Select a mathematical description or a representation that describes a problem

📌 **Chosen representation:**

- Simple arithmetic division
- No need for graphs or advanced models

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10. Item 3

Select a representation appropriate to the problem context

📌 **Lower relevance because:**

- The problem strongly cues the use of an average
- There is little need to choose among multiple representations

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11. Items 10 and 11 (Least Relevant)

Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

📌 **Reason for low relevance:**

- No diagrams or symbolic modelling are required
- The calculation can be completed with basic arithmetic

① 第 4 条（最核心）

Read, decode and make sense of statements … to create a model of the situation

📌 **为什么最重要?**
这道题首先考查的是:

- 能否读懂"only open from 1 July to 27 August"
- 能否理解"on average about how many people each day"

📌 **具体体现:**

- 需要把**文字信息**转化为一个可计算的情境
- 明白"平均每天"意味着 **总人数 ÷ 总天数**
- 不涉及复杂背景，但必须准确解读时间范围

➡️ **没有正确理解文本，就无法建模**

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② 第 6 条

Identify and describe the mathematical aspects of a real-world problem situation

📌 **体现在哪?**

- 这是一个真实世界的问题（游客爬富士山）
- 数学本质是:
  - 人数（200 000）
  - 时间（若干天）
  - 平均值

📌 **学生需要做到:**

- 抽取出"人数""天数""平均"这三个关键数学要素
- 忽略与计算无关的背景（例如富士山本身）

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③ 第 2 条

Identify the key variables in a model

📌 **关键变量非常清晰:**

- 总人数: $N = 200\,000$
- 天数: $T = 7/1$ 到 $8/27$ 的天数
- 日均人数: $\frac{N}{T}$

📌 **这一步是从情境走向模型的关键桥梁**

④ 第 7 条

Simplify or decompose a situation … to make it amenable to mathematical analysis

📌 **题目中的简化假设：**

- 假设每天人数**大致均匀**
- 忽略天气、周末、节假日等现实波动

📌 **学生要做的简化：**

- 不去纠结每天的变化
- 只保留"总量 ÷ 时间"的核心结构

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⑤ 第 12 条

Identify the constraints, assumptions, simplifications in a mathematical model

📌 **隐含假设：**

- 每天人数相同（平均意义）
- 登山季每天都开放
- 200 000 是一个近似值（about）

📌 **这是理解"average"概念的重要一环**

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⑥ 第 9 条

Translate a problem into a standard mathematical representation or algorithm

📌 **转换结果：**

$$\text{平均每天人数} = \frac{200\,000}{\text{天数}}$$

📌 **这是从语言到公式的直接翻译**

⑦ 第 8 条

Recognise aspects of a problem that correspond with known mathematical concepts

📌 **对应的数学知识：**

- 平均数（mean）
- 比率（rate）
- 单位时间内的数量

📌 **这是一个非常标准的"平均速度/平均人数"类问题**

⑧ 第 5 条

Recognise mathematical structure (relationships, patterns)

📌 **结构识别：**

- “总量—时间—单位量”的线性关系
- 人数与天数成反比

📌 **虽然结构简单，但依然存在**

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⑨ 第 1 条

Select a mathematical description or representation

📌 **选择了：**

- 算术除法模型
- 而不是图表、方程组等

📌 **难度较低，但仍有选择意识**

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⑩ 第 3 条

Select a representation appropriate to the problem context

📌 **为什么相关性较低？**

- 题目已经“暗示”使用平均值计算
- 不需要在多种表示（表格、图像）中做权衡

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⑪ 第 10 / 11 条（相关性最低）

Use mathematical tools (symbols, diagrams, variables)

📌 **原因：**

- 不需要图形、变量建模
- 简单心算或笔算即可完成

📌 **只涉及最基础的数值运算**